CYBERSECURITY / MACHINE LEARNING

PHISHGUARD AI

Machine Learning-Based Phishing Email Detection

An explainable educational prototype for classifying email text as **phishing** or **safe** using supervised binary classification.

17,513	98.12%	97.86%
Cleaned emails	Baseline accuracy	Baseline phishing recall

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Tech stack Python · Streamlit · pandas · scikit-learn

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Evidence note: The headline metrics are from the earlier evaluated model run. The updated dual-TF-IDF model has not been independently measured in the materials provided.

01 Project overview

Problem. Phishing emails impersonate trusted organizations, pressure users into disclosing secrets, and may blend into ordinary business messages. Manual review is slow; a text classifier can provide a preliminary triage signal.

Purpose. Build a local, interpretable phishing-email detector and a browser-based interface to evaluate message text. The system is meant to support learning and defensive experimentation, not act as a production email security gateway.

Project objectives

- Prepare a labeled email dataset with two classes: Safe (0) and Phishing (1).
- Train a supervised binary classifier with meaningful textual features.
- Evaluate on a held-out test set using Accuracy, Precision, Recall, F1 and a confusion matrix.
- Provide an interface where users can paste an email and view a predicted class and estimated phishing score.
- Inspect failures on difficult examples and identify ways to improve generalization.

System workflow

INPUT	PROCESSING	OUTPUT
Email subject and body pasted in Streamlit	Word and character TF-IDF → Logistic Regression	Predicted class: Safe / Phishing
Labeled training email corpus	Cleaning → split → fit → validation	Accuracy, Precision, Recall, F1, confusion matrix

Scope and boundaries

Included	Not included in the current implementation
Email text classification; binary labels; local training; UI; performance monitoring	Sender identity verification, DNS/domain reputation, URL destination analysis, attachment scanning, live inbox integration

Security interpretation: A “Safe” classification means the text scored below the chosen decision threshold; it does **not** guarantee that the email is trustworthy.

02 Data & preprocessing

Dataset: zefang-liu/phishing-email-dataset (Hugging Face), used as a CSV. The original dataset page displayed approximately 18,650 rows. The evaluated run displayed **17,513 cleaned usable emails** after the project filtering and duplicate handling.

Input fields and labels

CSV field	Role	Transformation
Email Text	Input feature (X)	Fill missing values, convert to string, trim whitespace
Email Type	Original category	Normalize case and spaces; map Safe Email to 0 and Phishing Email to 1
label	Target (y)	Discard unrecognized / null target values

Quality controls used in code

- Remove blank email messages and rows with unmapped labels.
- Normalize a text key (case and whitespace) to identify duplicates.
- Discard texts that have conflicting labels and keep one instance of each remaining exact normalized text.
- Use stratified splitting to preserve approximate class proportions.

Data partitioning

Partition	Share of total	Purpose
Fit / training (X_fit, y_fit)	64%	Learn TF-IDF vocabulary and classifier weights
Validation (X_val, y_val)	16%	Choose the decision threshold
Final test (X_test, y_test)	20%	Estimate performance after design choices are fixed

A first split reserves 20% for testing. A second split reserves 20% of the 80% development portion for validation. Both splits use **random_state=42** and stratification. The hold-out test must not guide repeated tuning; otherwise its independence is reduced.

03 Model design

Feature extraction: two complementary views of text

Component	Configuration	Function
Word TF-IDF	max_features=20,000; ngram_range=(1,2); min_df=2; sublinear_tf=True	Captures informative words and adjacent word pairs.
Character TF-IDF	analyzer=char_wb; ngram_range=(3,4); max_features=10,000; min_df=2	Captures short character sequences and spelling patterns.
FeatureUnion	Word features + character features	Concatenates both feature spaces into one matrix.

Classifier

Parameter	Setting	Purpose
Algorithm	Logistic Regression	Supervised binary text classification
max_iter	1000	Iteration limit for solver convergence
class_weight	balanced	Compensates for uneven sample frequencies
solver	liblinear	Optimization method
random_state	42	Reproducibility

Validation-based threshold selection

The fitted classifier produces a phishing score for each validation message. The application tests candidate thresholds of **0.35, 0.40, ... , 0.75** (step 0.05). It keeps thresholds whose validation Recall is at least **0.95**, then picks the highest validation F1. A tie favors the candidate closest to 0.50.

Fallback behavior: If no candidate achieves 95% validation recall, the code retains threshold 0.50. Therefore 95% recall is a **selection target**, not a guarantee. The selected threshold is stored as **model.threshold_**.

Prediction rule: A message is predicted Phishing (1) when its model score is greater than or equal to the selected threshold; otherwise it is predicted Safe (0).

04 Evaluation evidence

The dashboard screenshot supplied for the **earlier evaluated implementation** reported the following held-out test metrics. These numbers should not be presented as verified results for the later revised word-plus-character model until that version is rerun.

98.12%	97.11%	97.86%	97.48%
Accuracy	Precision	Recall	F1 score

Confusion matrix: 3,503 test messages

Actual \ Predicted	Safe	Phishing
Safe (actual)	2,158 - True negatives	38 - False positives
Phishing (actual)	28 - False negatives	1,279 - True positives

Metric / count	Interpretation
Correct decisions: 3,437 / 3,503	Most messages were correctly labeled in the held-out sample.
False positives: 38	Legitimate email texts flagged as phishing.
False negatives: 28	Phishing email texts missed, a security-relevant failure mode.
Preprocessing output: 17,513 rows	Number displayed as “total emails used” in that run.

How to interpret the results

Accuracy summarizes correctness across both classes. Precision answers: “Of the emails flagged as phishing, how many truly were phishing?” Recall answers: “Of the phishing emails, how many were caught?” F1 is the harmonic mean of Precision and Recall.

Evidence limitation: Dataset-based results can overstate performance on unseen campaigns, languages, organizations and contemporary messages. Assess with independent, representative examples and avoid optimizing repeatedly against the final hold-out.

05 Manual challenge tests

The user tested eight synthetic messages with the earlier application. These are useful error probes, but they are a very small, hand-written sample - not a statistically reliable estimate of production accuracy.

4 / 8	3	1	50%
Correct	False positives	False negative	Manual accuracy

Observed outcomes

# Scenario	Expected	Observed	Outcome
1 Account verification	Phishing	Phishing	Correct
2 Team meeting	Safe	Safe	Correct
3 Prize notification	Phishing	Phishing	Correct
4 Password reset confirmation	Safe	Phishing	False positive
5 Fake IT support	Phishing	Phishing	Correct
6 Shopping receipt	Safe	Phishing	False positive
7 Shared document sign-in	Phishing	Safe	False negative
8 Security awareness notice	Safe	Phishing	False positive

Key findings

Over-alerting: Legitimate messages about accounts, password changes, or transactions may use terminology associated with phishing in the training set. This can generate false positives.

Under-detection: Socially plausible document-sharing lures may lack obvious phishing terms and escape text-only detection. For the synthetic case, the malicious intent is assumed by the test scenario; it cannot be confirmed without inspecting the sender and URL.

Manual Recall = 75% (3 of 4 phishing cases detected); **Manual Precision = 50%** (3 of 6 phishing predictions correct); **Manual F1 = 60%**. These values apply only to the eight scenarios.

06 Implementation & next steps

Project structure

File / directory	Role
work&train/app.py	Streamlit interface: profile links, scanner, model metrics
work&train/model.py	Data cleaning, feature extraction, fitting, threshold tuning, prediction
work&train/style.css	Colors, layout and profile-image styles
work&train/profile.jpg (optional)	Developer image shown in the application header
csvdata/*.csv	Email text and labels; set / discover the actual filename in app.py
requirements.txt	Installable Python dependency list (provided separately)

Local setup

1. Install Python 3.13 (the provided environment used Python 3.13).
2. Open a terminal in the project directory.
3. Install dependencies using **py -m pip install -r requirements.txt**.
4. Confirm that the dataset CSV exists and matches the path in app.py.
5. Enter the folder containing app.py and start **py -m streamlit run app.py**.
6. Open the local URL printed by Streamlit (usually http://localhost:8501).

Suggested roadmap

- Re-evaluate the revised dual-TF-IDF model and record a fresh confusion matrix; compare it fairly with the baseline.
- Track false positives and false negatives on an independent set of emails; capture model scores and actual labels.
- Explore header authentication (SPF / DKIM / DMARC), sender/domain analysis, and URL features as separate, carefully validated signals.
- Save the trained model and threshold using an appropriate local artifact format to avoid training on each launch.
- Consider calibration, multilingual coverage and privacy constraints before any real-world deployment.

Sources and project links

Dataset: huggingface.co/datasets/zefang-liu/phishing-email-dataset
GitHub profile: github.com/Moataz125
LinkedIn profile: linkedin.com/in/moataz-farhan-60628830a

RESPONSIBLE USE | Educational text screening only. A Safe prediction does not verify sender identity, URLs, or attachments.

APPENDIX A / EMAIL TEST SCENARIOS

14 synthetic examples for manual testing of the local phishing detector.

Record the model prediction and phishing score; these expected labels are scenario assumptions.

01 Fake password expiration

EXPECTED: PHISHING

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Password Expiration Notice

Hello,

Your company mailbox will be disabled today unless you verify your account.

Open https://mail-confirmation.example/login and enter your email password and one-time verification code.

Mailbox Support

WHY THIS LABEL IS EXPECTED

Urgency, off-platform login URL, and a request for password and OTP.

MODEL PREDICTION: _____ PHISHING SCORE: _____

02 Password change confirmation

EXPECTED: SAFE

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Your Password Was Changed

Hello Alex,

This email confirms that your account password was changed at 10:15 AM.

If you made this change, no action is required. Otherwise, open the official app directly and contact support from the Help menu.

Account Security

WHY THIS LABEL IS EXPECTED

Security vocabulary is present, but no secrets or external sign-in are requested.

MODEL PREDICTION: _____ PHISHING SCORE: _____

APPENDIX A / EMAIL TEST SCENARIOS

Copy each complete message, including the subject, into PhishGuard AI.

03 Weekly team meeting

EXPECTED: SAFE

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Weekly Project Meeting

Hi team,

Our weekly project meeting is on Tuesday at 11:00 AM in Meeting Room 2. Please bring your progress notes and questions.

Thanks,
Project Coordinator

WHY THIS LABEL IS EXPECTED

Normal operational message with no suspicious request.

MODEL PREDICTION: _____ PHISHING SCORE: _____

04 CEO gift card request

EXPECTED: PHISHING

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Quick Confidential Favor

Hi Emma,

I am in meetings and need you to purchase six gift cards for an urgent client thank-you. Keep this confidential until I am finished. Send the card codes to me by replying to this email.

Thanks,
CEO

WHY THIS LABEL IS EXPECTED

Business email compromise scenario: secret, urgent purchase of transferable value.

MODEL PREDICTION: _____ PHISHING SCORE: _____

APPENDIX A / EMAIL TEST SCENARIOS

Copy each complete message, including the subject, into PhishGuard AI.

05 Order shipping receipt

EXPECTED: SAFE

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Your Order Has Shipped

Hi Sam,

Your order #ORD-82431 has shipped. The delivery window is Friday between 10 AM and 2 PM.

For details, open the shopping app directly and view your order history.

Customer Care

WHY THIS LABEL IS EXPECTED

Routine shipping update without external payment or credential request.

MODEL PREDICTION: _____ PHISHING SCORE: _____

06 Fake delivery fee

EXPECTED: PHISHING

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Delivery Held - Action Needed

Dear Customer,

Your parcel is on hold because a small customs fee is unpaid. Pay the fee in the next 2 hours at https://parcel-release.example/payment to avoid package return.

Delivery Service

WHY THIS LABEL IS EXPECTED

Urgency and unverified payment request through an external URL.

MODEL PREDICTION: _____ PHISHING SCORE: _____

APPENDIX A / EMAIL TEST SCENARIOS

Copy each complete message, including the subject, into PhishGuard AI.

07 MFA security reminder

EXPECTED: SAFE

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Multi-Factor Authentication Reminder

Hello employees,

Never share an authentication code or password with anyone, including IT staff. If you receive a suspicious sign-in prompt, deny it and report the event through the official internal support portal.

Security Awareness Team

WHY THIS LABEL IS EXPECTED

Legitimate security guidance; it warns against sharing codes.

MODEL PREDICTION: _____ PHISHING SCORE: _____

08 Fake IT verification

EXPECTED: PHISHING

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: IT Account Review

Hello employee,

The help desk is verifying user accounts today. Reply with your corporate username, current password, and the six-digit sign-in code from your phone.

IT Operations

WHY THIS LABEL IS EXPECTED

An impersonated IT request for credentials and a second factor.

MODEL PREDICTION: _____ PHISHING SCORE: _____

APPENDIX A / EMAIL TEST SCENARIOS

Copy each complete message, including the subject, into PhishGuard AI.

09 Bank account change invoice

EXPECTED: PHISHING

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Updated Bank Details for Invoice 4472

Hi Finance,

Please pay the outstanding vendor invoice using our new bank account today. The old account has been closed, and this update is confidential. Skip the normal phone verification because the manager already approved it.

Vendor Billing

WHY THIS LABEL IS EXPECTED

Business email compromise pattern: payment redirection and bypass of verification.

MODEL PREDICTION: _____ PHISHING SCORE: _____

10 Internal document review

EXPECTED: SAFE

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Design Review Document

Hi Alex,

I made a few edits to the design notes we discussed. The updated version is available in the team's usual workspace. Open the official company portal directly and review it before Friday.

Thanks,
Daniel

WHY THIS LABEL IS EXPECTED

A document review request that directs users to the usual company portal.

MODEL PREDICTION: _____ PHISHING SCORE: _____

APPENDIX A / EMAIL TEST SCENARIOS

Copy each complete message, including the subject, into PhishGuard AI.

11 External document login lure

EXPECTED: PHISHING

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Revised Project Budget

Hi Alex,

Here is the revised budget from this morning's meeting. Please open https://shared-budget-review.example/signin and type your company password to unlock the file before our call.

Regards,
Michael

WHY THIS LABEL IS EXPECTED

Synthetic phishing scenario assuming the external sign-in page is not authorized; text alone cannot verify the domain.

MODEL PREDICTION: _____ PHISHING SCORE: _____

12 Security bulletin

EXPECTED: SAFE

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Increase in Phishing Attempts

Dear colleagues,

Our security team has observed more phishing emails using the words urgent, password and account locked. Do not click unexpected links. Report suspicious messages using the Report Phishing option in your email client.

Information Security

WHY THIS LABEL IS EXPECTED

Defensive awareness email with many phishing-related terms.

MODEL PREDICTION: _____ PHISHING SCORE: _____

APPENDIX A / EMAIL TEST SCENARIOS

Copy each complete message, including the subject, into PhishGuard AI.

13 Fake university scholarship

EXPECTED: PHISHING

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Scholarship Award Confirmation

Dear Student,

You have been selected for a tuition grant. To receive the funds, submit your student login password, national ID scan, and bank card details by the end of today.

Scholarship Office

WHY THIS LABEL IS EXPECTED

Unsolicited award, deadline pressure, and collection of sensitive information.

MODEL PREDICTION: _____ PHISHING SCORE: _____

14 Lecture reschedule

EXPECTED: SAFE

EMAIL MESSAGE / COPY SUBJECT + BODY INTO THE SCANNER

Subject: Cybersecurity Lecture Update

Dear students,

Tomorrow's lecture will start at 10:30 AM instead of 9:00 AM. The classroom remains B214. The assignment deadline is unchanged.

Best regards,
Course Instructor

WHY THIS LABEL IS EXPECTED

Routine academic scheduling information.

MODEL PREDICTION: _____ PHISHING SCORE: _____